CHAPTER III

RESEARCH METHODOLOGY

This chapter outlines the research methodology employed in this study, detailing the approaches and procedures used to gather and analyze data.

Research Design

This section states the research design which refers to the overall strategy that you choose to integrate the different components of the study in a coherent and logical way, thereby, ensuring you will effectively address the research problem; it constitutes the blueprint for the collection, measurement, and analysis of data.

Note that your research problem determines the type of design you should use, not the other way around! *(*De Vaus, D. A. Research Design in Social Research. London: SAGE, 2001; Trochim, William M.K. Research Methods Knowledge Base. 2006.) Research design for computer science can be a combination Experimental study which focuses of the trials of various algorithms that would best perform its purpose while Development and Application studies for the proof of concept and prototyping of the software.

Explain the research design you will use. Discuss your reasoning for choosing either qualitative or quantitative research or for using a blended approach or descriptive, correlational, experimental, developmental, or any alternative method your specific field recognizes (Refer to Module #1 to check what research design is appropriate to your study).

EXAMPLE. This study adopts a hybrid research design combining development and experimental approaches. The development component focuses on creating a predictive analytics website that utilizes machine learning algorithms to forecast user engagement, while the experimental component evaluates the system's accuracy and performance through trial runs. This design ensures a structured process for building the system while empirically assessing its effectiveness in meeting the research objectives.

Locale of the Study

This section discusses the target locale for this research. Also, the nature (who, where, what) of the selected locale where the researchers will conduct a research study, this should be based on the objectives to be achieved.

EXAMPLE. The research will be conducted at Company XYZ, a retail enterprise located in the city of ABC. The target data will be collected from the company's e-commerce platform, which serves a diverse customer base within the city and surrounding areas. This setting provides a real-world environment for analyzing user interactions and implementing predictive analytics to enhance customer engagement and sales strategies. The target locale includes users accessing the website from different demographics to ensure diverse data collection. The focus is on analyzing user behavior and predicting future engagement to improve personalization strategies.

Applied Concepts and Techniques

This section states the applied concepts and techniques to be used in solving the research problem and achieving the research objectives. Indicate theoretical disciplines computer science such as algorithms, theory of computation, information theory, and automation or the to the data-driven disciplines and computational process such as data science, data mining, deep learning, machine learning or Artificial Intelligence. Specifically discuss the theoretical background align to the research area focus. For example: Machine Learning → Unsupervised Learning → Clustering → K Means Clustering/ hierarchical clustering/ Density-Based Spatial Clustering of Applications with Noise/ Gaussian Mixtures Model.

EXAMPLE. The core concepts underpinning this research involve machine learning, data mining, and predictive analytics. Specifically, supervised learning techniques such as decision trees and support vector machines will be employed to develop predictive models. These concepts are grounded in data science theories related to classification algorithms and statistical modeling, providing the foundation for accurate user engagement predictions.

Algorithm Analysis

This section states the methodology, or the techniques used in data or algorithm analysis. An algorithm analysis is a technique that is used to measure the performance of the algorithms.

Speed is one of the key parameters in determining the potential of an algorithm. Discuss the complexity and metrics used to measure parameters. There are some other factors like performance, accuracy, time and space

complexity and other metrics that determine the quality of an algorithm. This section must also cover the criteria to be satisfied by the algorithm such as input, output, definiteness, finiteness, and effectiveness. For instance, in machine learning or deep learning model we check the performance in terms of: confusion matrix, accuracy, precision, recall, specificity, F1 score, precision-Recall or Receiver Operating Characteristics (ROC) curve.

For a study with emphasis on hardware, you may introduce your method on analyzing the system. You may consider Multiple-Criteria Decision Analysis (MCDA), also known as Multiple-Criteria Decision-Making (MCDM), is a structured decision-making approach that helps individuals and groups choose among alternatives by considering multiple, often conflicting, quantitative and qualitative criteria. It involves identifying options, defining criteria, assigning weights to them, scoring how well each option performs on each criterion, and then using these scores to rank the options and select the best solution.

EXAMPLE. The algorithms implemented will be analyzed based on their accuracy, precision, recall, and F1-score. Computational complexity will be measured using time and space metrics, ensuring that the models are optimized for real-time predictions. Criteria such as input data quality, model scalability, and interpretability will also be considered to evaluate algorithm performance effectively.

The performance of the algorithms will be evaluated using several key metrics. The accuracy (ACC) is calculated as the ratio of correctly predicted instances to the total number of instances, expressed as:

$$Accuracy = (TP + TN) / (TP + FP + TN + FN)$$

where TP (True Positives) and TN (True Negatives) represent correctly classified positive and negative instances, respectively, while FP (False Positives) and FN (False Negatives) denote misclassifications.

Precision (P) measures the proportion of positive identifications that were actually correct:

$$Precision = TP / (TP + FP)$$

Recall (R), also known as sensitivity, indicates the proportion of actual positives correctly identified:

$$Recall = TP / (TP + FN)$$

The F1 score is the harmonic mean of precision and recall, providing a balanced measure:

$$F1 = 2 * (Precision * Recall) / (Precision + Recall)$$

Additionally, the Receiver Operating Characteristic (ROC) curve's Area Under the Curve (AUC) will be used to evaluate the model's overall ability to distinguish between classes.

The computational complexity of the algorithms will be analyzed based on their time complexity O(n) and space complexity O(n), where n denotes the number of data points processed.

Data Collection Methods

This section states the data source and nature of the data, present the data sets used in a table with its attribute, description, and values.

In the Data Collection Methods section, clearly specify the sources of your data, whether they are primary or secondary, and describe how the data will be gathered. Include details about the nature and attributes of the data, such as the type of data (quantitative or qualitative), the format (structured or unstructured), and the specific variables or features relevant to your study. Outline the data collection process, including tools, techniques, or instruments used (e.g., web scraping, surveys, logs), and the timeframe for data collection. Additionally, provide a table or list summarizing the datasets, attributes, descriptions, and sample values to offer clarity and transparency on the data sources and their relevance to your research objectives.

EXAMPLE. Data will be collected from the server logs of the target e-commerce website over a period of three months. The dataset includes attributes such as user ID, session duration, pages visited, and purchase history. Table 3 summarizes the dataset attributes, descriptions, and sample values to provide clarity on data sources and structure.

Table 3. Dataset Attributes Description

Attribute	Description	Sample Values
User ID	Unique identifier for users	U12345
Session Duration	Time spent per session (seconds)	300, 450
Pages Visited	Number of pages visited	5, 8, 12
Purchase History	Number of purchases made	0, 1, 2

Data Model Generation

This section states the framework in generating data model as stated on the objectives of the study. Elaborate the phases of the selected CS solutions. In

data mining, CRISP-DM to KDD is usually used enumerate the phases and discuss your techniques in achieving every tasks. In ML Process there are several frameworks which can be generally broken into several steps: such as Collection of Data, Data Wrangling, Model Building, Model Evaluation and Model Deployment. Please ensure that all methods, techniques, and approach will be based on scientific process and reputable references.

EXAMPLE.

The data modeling process follows the CRISP-DM framework, beginning with data collection and cleaning, followed by feature engineering, and then model training using supervised learning algorithms. Model evaluation will be conducted through cross-validation and performance metrics such as accuracy and ROC-AUC. The final model will be integrated into the website to enable real-time predictive analytics, aligning with the study's objectives.

Data Collection and Cleaning

Data will be gathered from server logs of the e-commerce website over a three-month period, capturing user interactions such as page visits and purchase history. The collected data will then undergo cleaning to remove duplicates, fill missing values, and standardize formats to ensure quality and consistency.

Feature Engineering

Relevant features will be extracted and transformed from the raw data, such as calculating session duration and aggregating user activity metrics. These features will be normalized and encoded to prepare them for input into the machine learning models, enhancing predictive accuracy.

Model Building

Supervised learning algorithms like decision trees and support vector machines will be trained using the processed dataset to predict user engagement levels. Hyperparameter tuning will be performed to optimize model performance and prevent overfitting.

Model Evaluation

The models will be evaluated using cross-validation techniques and metrics such as accuracy, precision, recall, and ROC-AUC to assess their effectiveness. The best-performing model will be selected based on these evaluation results for deployment in the website system.

Model Deployment

The finalized predictive model will be integrated into the website's backend, enabling real-time user engagement predictions during interactions. Continuous monitoring will be implemented to track model performance and facilitate updates as new data becomes available.

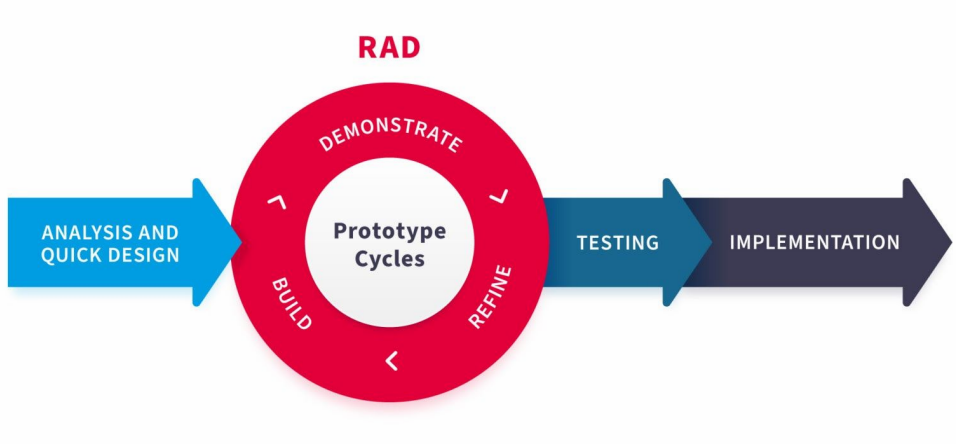
System Development Methodology

This section discusses the researchers' developmental methodology as the algorithm or data model integration in the software design. The models specify the various stages of the process and the order in which they are carried out. Software

process model may also be reflected on this section.

Describe what Software Development Life Cycle (SDLC) you will follow. E.g. waterfall methodology, iterative model, continuous model, agile

methodology, lean development, rapid application development, prototype model, feature driven, scrum, or any alternative method you know.



Source: Website Name/Article Name (if any)

Figure 2. Rapid Application Development

You may add figures in your discussion. Label of tables and figures are centered, bold and size 10. For tables, the label must be placed above the table while for figures it must be place below the figure. All figures must be cited within the text. The word Figure must be used instead of figure or Fig when referring to the Figure. If figures are small, they can be arranged differently.

All tables must be cited within the text. For example, Table 1 should be referred to as Table 1 and not table 1 or anything else. Font size for text within tables should normally be at 10 pt. If majority of the tables are small in width, they can be left-aligned but this must be consistent throughout the paper.

Explain what actions you will take on each phase of the model. If your study requires you to add system/project design, you may add discussion on models such as Use Case Diagram, Context Diagram, Data Flow Diagram, or

any other diagrams/models related and will support the discussion of your methodology.

EXAMPLE.

The system development employs an iterative Agile methodology, facilitating continuous testing and refinement of the predictive model within the website interface. The phases include requirements analysis, system design, implementation, testing, and deployment. This approach ensures flexibility and responsiveness to any emerging challenges during development.

Requirements Analysis

The requirements analysis phase involves gathering and defining the features needed for the predictive analytics website, focusing on user engagement metrics and system functionalities. This ensures that all stakeholder needs are accurately captured to guide the development process.

System Design

During the system design phase, the architecture of the website and the predictive model are planned, including data flow, user interface layout, and algorithm selection. This phase establishes a clear blueprint to facilitate effective implementation and integration.

Implementation

The implementation phase involves coding the web application, integrating machine learning models, and setting up data pipelines using tools like Python and Flask. Developers work iteratively to build functional modules that align with the designed architecture.

Testing

Testing includes conducting unit tests on individual components, system tests for overall functionality, and user acceptance tests to ensure usability and accuracy. These activities help identify and resolve issues early, ensuring a reliable and effective system.

Deployment

The final system is deployed onto a live server, making it accessible to users and enabling real-time data processing and predictions. Post-deployment monitoring ensures system stability and facilitates future updates based on user feedback.

Software Tools Used

This section discusses the programming language and software tools used in data modelling and system development. Elaborate the specific tasks and uses of each tool.

Scikit-learn Library. Software used to build machine learning models. This software allows to create and construct a library of models at the preferred level of abstraction to provide information/results for a particular function.

Python Programming Language. The programming language used in developing and implementing the developed machine learning in the web-based system.

Hardware Requirements

This section outlines the hardware components necessary for data modeling and system development. Discuss the specifications and configurations

required throughout development and deployment phases. Elaborate the specific tasks and uses of each hardware.

Desktop. It is used to perform the development of mobile application software. It must have a good specification which can run the used software such as Visual Studio Code and TensorFlow. Only one computer is used during the development of the said web based.

Table 4. Desktop Minimum Specification

Processor	Processor Intel ® core ™ i3-4030 CPU @ 3.30GHz
Memory	2.00 Gigabytes
Hard Disk Drive	10 Gigabytes
System Type	32-bit Operating System

Android Phone. It is the most important hardware device for the implementation of the project. Developing a responsive web-based system will help everyone to access the recommender system through their mobile phone without using a high-end device.

Table 5. Android Phone Minimum Specification

Processor	1.95 GHz Snapdragon 439 Octa core
Memory	2 GB Ram
Internal Storage	2 Gigabytes
Browser	Chrome

System Architecture

This section presents the system architecture of the developed system. Enumerate the functions and features highlight the integration of computing science solutions. Discuss the structure, behavior, and more views of a system.

Identify and describe the main components or modules of the system, such as data sources, processing units, analytical engines, and user interfaces. Explain how these components interact with each other, illustrating the flow of data and control within the system. Use diagrams or schematics if necessary to enhance clarity and visual understanding.

To enhance understanding, incorporate relevant diagrams such as Entity-Relationship (ER) diagrams, Unified Modeling Language (UML) diagrams, Data Flow Diagrams (DFD), or system block diagrams. These visual tools help illustrate the architecture's structure, data movement, and component interactions effectively. You may also choose other diagram types like architecture diagrams, sequence diagrams, or component diagrams based on what best represents your system’s design.

EXAMPLE.

The system architecture comprises three main components: the data ingestion module, the predictive analytics engine, and the user interface. The architecture employs a client-server model, with the server hosting the predictive models and the web interface. Data flows from user interactions to the server, which processes the data, applies the predictive model, and displays personalized recommendations or predictions to users.

Software Testing

This section discusses the software testing method and strategies to determine if the computing science problem was solved.

To write the Software Testing section, begin by outlining the different testing strategies that will be employed, such as unit testing, integration testing, user acceptance testing, and performance testing. Describe the purpose of each strategy and emphasize the importance of verifying individual modules, system components, usability, and system performance. Include detailed procedures for each testing type, specifying how tests will be conducted, the tools or frameworks to be used, and the specific steps involved such as designing test cases, simulating data flow, and gathering user feedback.

If your system involves testing or evaluation with users, discuss the sampling design, including the technique (probability or non-probability), sample size calculation, and rationale. Describe the data collection instruments used, such as surveys or questionnaires, and detail their validation process for face and content validity if applicable. Outline the statistical treatments employed to analyze the data, such as techniques for identifying relationships, correlations, or differences among variables.

EXAMPLE.

Testing strategies include unit testing for individual modules, integration testing for system components, and user acceptance testing to ensure usability. Performance testing will assess the system's response time and prediction accuracy under different load conditions. These strategies guarantee that the

system effectively solves the predictive analytics problem with reliability and robustness.

Unit Testing Procedure

Each module, such as data preprocessing, feature extraction, and the predictive model, will undergo unit testing using frameworks like pytest. Test cases will be designed to verify that each function performs as expected with both typical and edge-case inputs. For example, the data cleaning function will be tested with datasets containing missing or inconsistent values to ensure robustness.

Integration Testing Procedure

Once individual modules pass unit tests, integration testing will be conducted to validate the interaction between components. This will involve simulating data flow from the user interface through the server to the predictive engine and back. Tools such as Postman or Selenium can be used to automate testing of API endpoints and user interactions, ensuring seamless communication and data exchange.

Performance Testing Procedure

Using load testing tools like JMeter, the system will be subjected to simulated concurrent users to evaluate response times and stability under stress. The system's prediction accuracy will also be monitored by processing a set of validation data repeatedly to ensure consistent performance. Metrics such as response time, throughput, and error rates will be recorded and analyzed to identify potential bottlenecks and optimize system performance.

User Acceptance Testing Procedure

A group of end-users (e.g., website administrators or selected users) will evaluate the system's usability and functionality. Users will perform common tasks, such as submitting data for prediction and interpreting results, while providing feedback on the interface's clarity and responsiveness. This stage ensures the system meets user expectations and operational requirements.

Sampling Design. For the user testing of the predictive analytics website, a purposive sampling method was employed to target users who have prior experience with similar systems. This approach ensured that participants possessed relevant knowledge and could provide meaningful feedback. A total of 30 participants were selected based on criteria such as familiarity with data analytics tools and active usage of online platforms. The sample size was calculated using Cochran’s formula to ensure a confidence level of 95% and a margin of error of 5%. This systematic approach helped in obtaining a representative sample that could reliably reflect the target user population.

Data Collection Instrument. The primary data collection tool was a structured online survey designed to gauge user satisfaction, ease of use, and perceived system accuracy. The questionnaire included Likert-scale items, multiple-choice questions, and open-ended responses to capture both quantitative and qualitative insights. Prior to deployment, the survey instrument was reviewed by subject matter experts to ensure clarity and relevance. A pilot test was conducted with five users similar to the target population to identify

ambiguities and improve question clarity. Based on feedback, minor revisions were made to enhance the instrument's reliability and validity.

Validation of the Questionnaire. The validation process involved assessing the questionnaire for face and content validity. Subject matter experts in data analytics and user experience reviewed the instrument to ensure that it accurately measured the intended constructs and covered all relevant aspects of system evaluation. Their feedback helped refine the questions for clarity, appropriateness, and comprehensiveness. The revised questionnaire was then pilot tested with a small sample to confirm its effectiveness, ensuring that it reliably captured user perceptions and system performance data.

Statistical Treatment. Collected data were analyzed using descriptive statistics such as mean scores, frequency distributions, and standard deviations to summarize user responses. To explore relationships between variables, such as user satisfaction and perceived system accuracy, Pearson correlation coefficients were calculated. Inferential statistics, including t-tests, were employed to determine significant differences between user groups. All statistical analyses were performed using SPSS software, ensuring the data's integrity and the validity of conclusions drawn about system usability and performance based on user feedback.